

Stock Portfolio Optimisation

Presented by Alpay, Rohan,
Byron, Tymon, Yalindee

The *Question.*



SCHOOL A • PREDICTION



Predict future returns better, get a better portfolio.

→ FORECASTING • ML • MEAN-VARIANCE

SCHOOL B • STRUCTURE



Don't predict, diversify and minimise the risk you can measure.

→ DIVERSIFICATION • RISK PARITY • HRP

vs

First, the *vocabulary*.



01

Stock

AAPL · MSFT · JNJ



02

Portfolio

N STOCKS · WEIGHTS W_i



03

Sharpe Ratio

$$S = (R_p - R_f) / \sigma_p$$



WHY WE OPTIMISE

Find the *best*
way to invest.



OBJECTIVE A

Maximise return



OBJECTIVE B

Minimise risk



TRADE - OFF

Efficient frontier

Data cleaning *pipeline*.

1,199 → 55
RAW INVESTABLE

01



Format & assess

Convert datetime, audit missing data, delisted stocks, single-day outliers.

02



Filter coverage

Drop stocks with under 50% coverage and extreme outliers.

03



Forward fill

Fill price gaps between non-trading periods with last observed price.

04



Winsorise

Cap extreme daily returns at $\pm 50\%$ to tame outliers.

05



Map sectors

Tag remaining tickers with 11 GICS sectors.

06



Handpick

Top 5 Sharpe-ratio stocks per sector → final 55.



Why this matters: a clean, investable universe is the precondition for every optimisation that follows.



SECTION
02

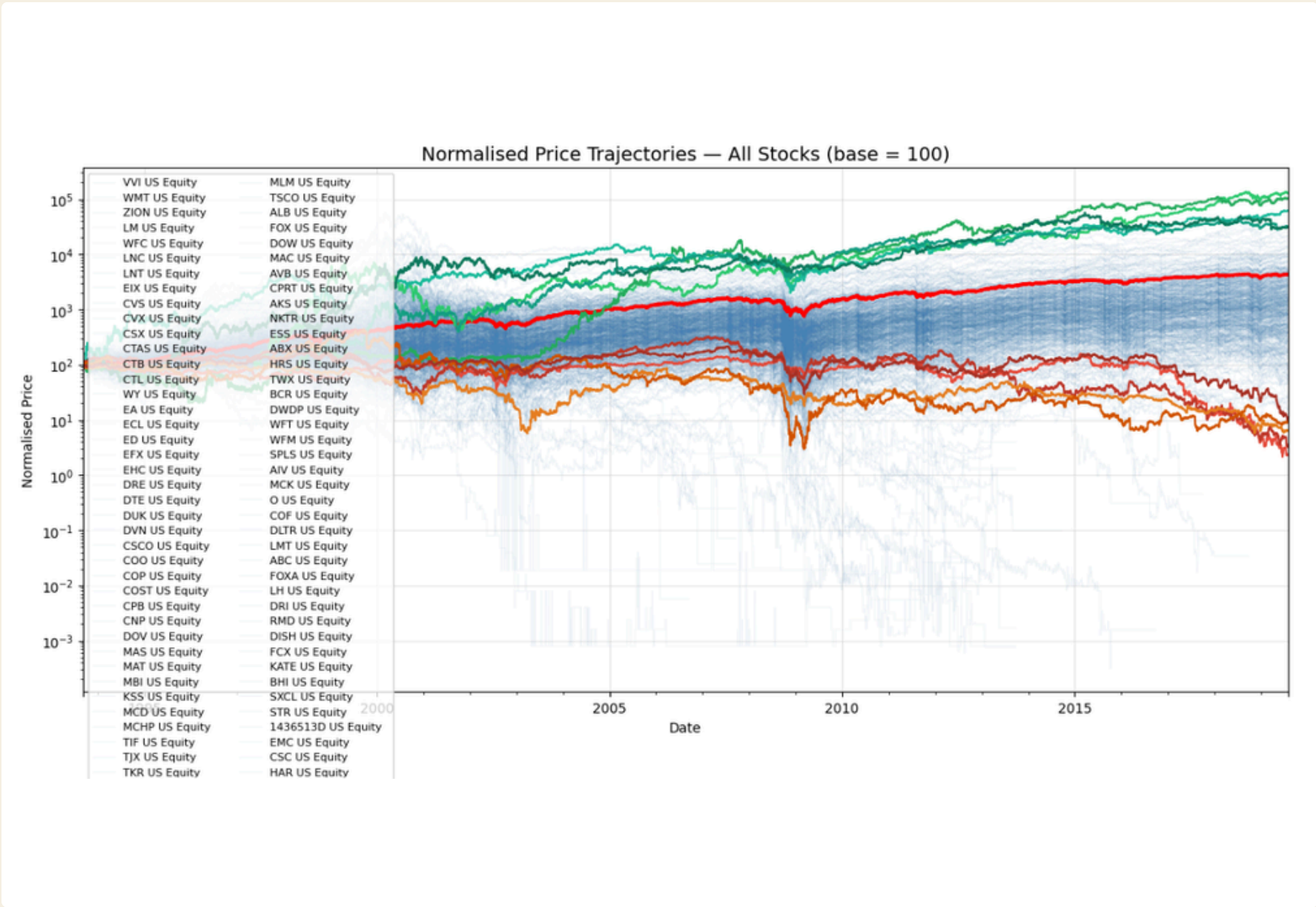
Exploratory *Data Analysis.*

Before optimising, a key understanding of the data is needed.

Exploratory analysis – *all stocks*.

FIG 01 · NORMALISED PRICE TRAJECTORIES

DIVERSIFICATION WORKS



Single stocks trail the **equal-weighted portfolio line** (red). Negative skew across trend lines: losses bigger than gains.

FINDING · LOW CORRELATION

0.277 avg pairwise
9 pairs show strong correlation.

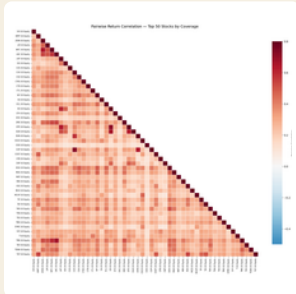
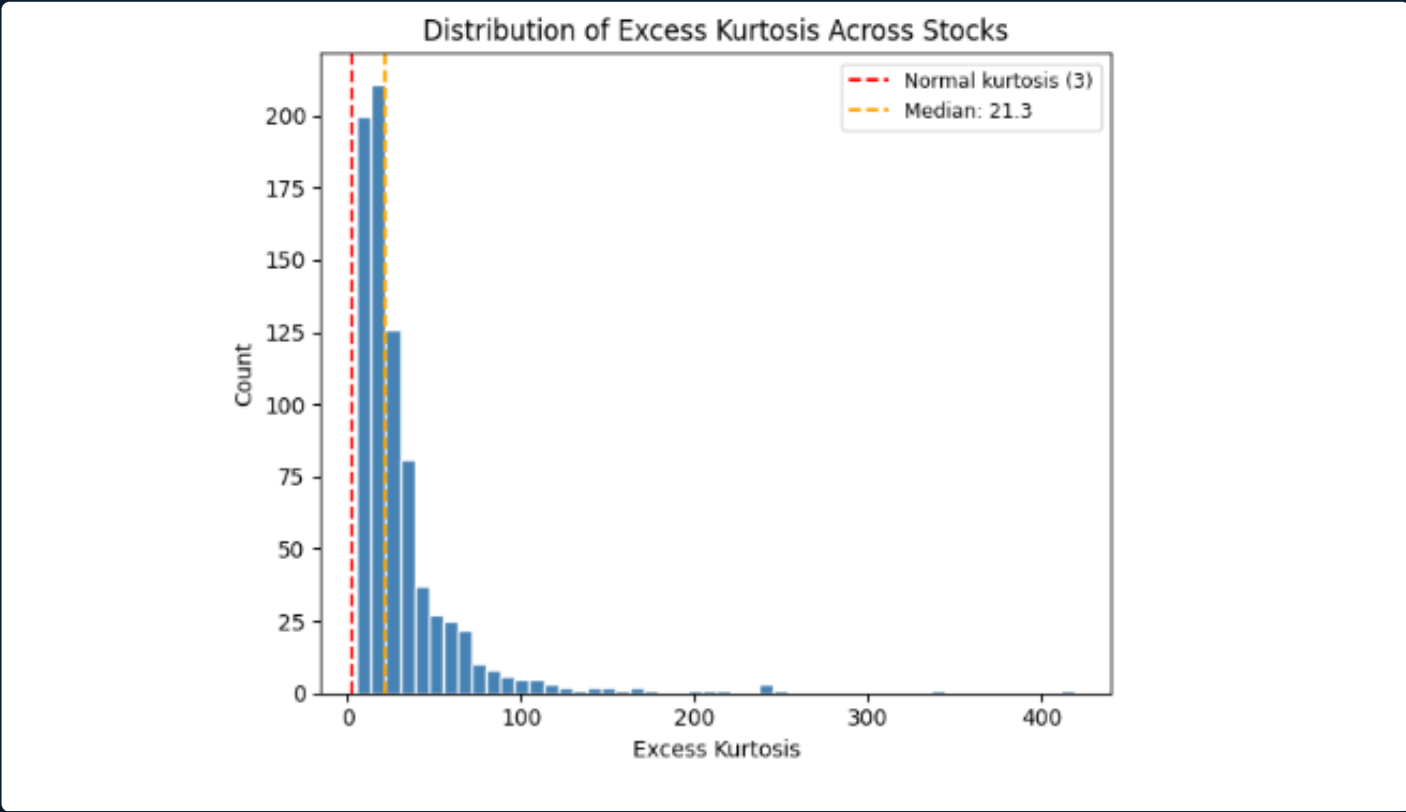


FIG 02 · EXCESS KURTOSIS DISTRIBUTION

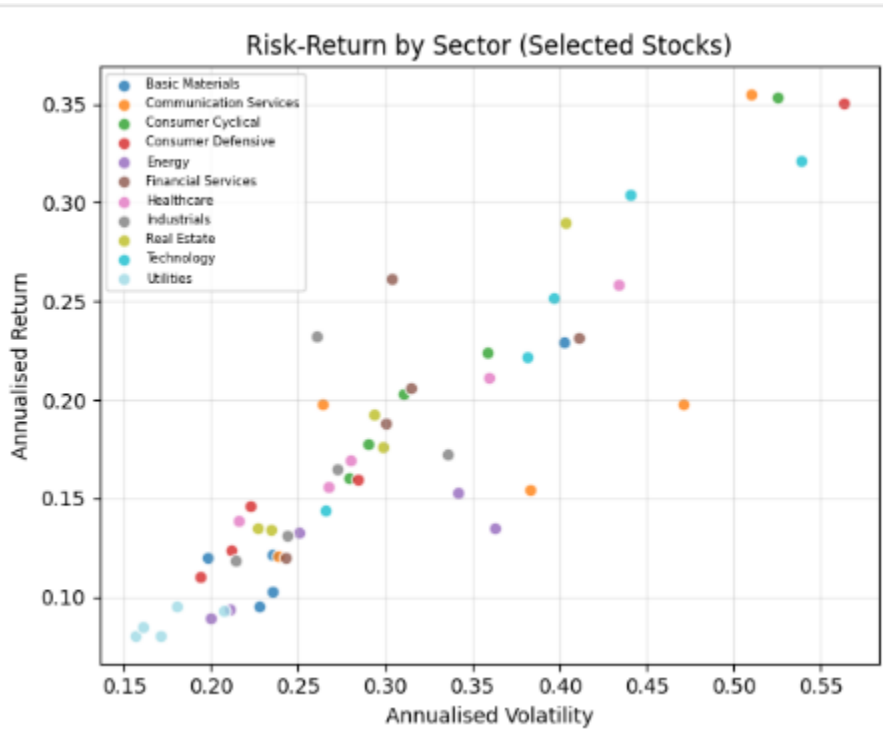
FAT TAILS



Median **21.3** vs Gaussian 3 — extreme moves are far more common than normal predicts.

Industry-specific *cuts*.

FIG 03 • RISK VS RETURN BY SECTOR



Industries hug the risk-return line. **Outliers** signal spurred performance, both upside and volatility.

TOP SHARPE



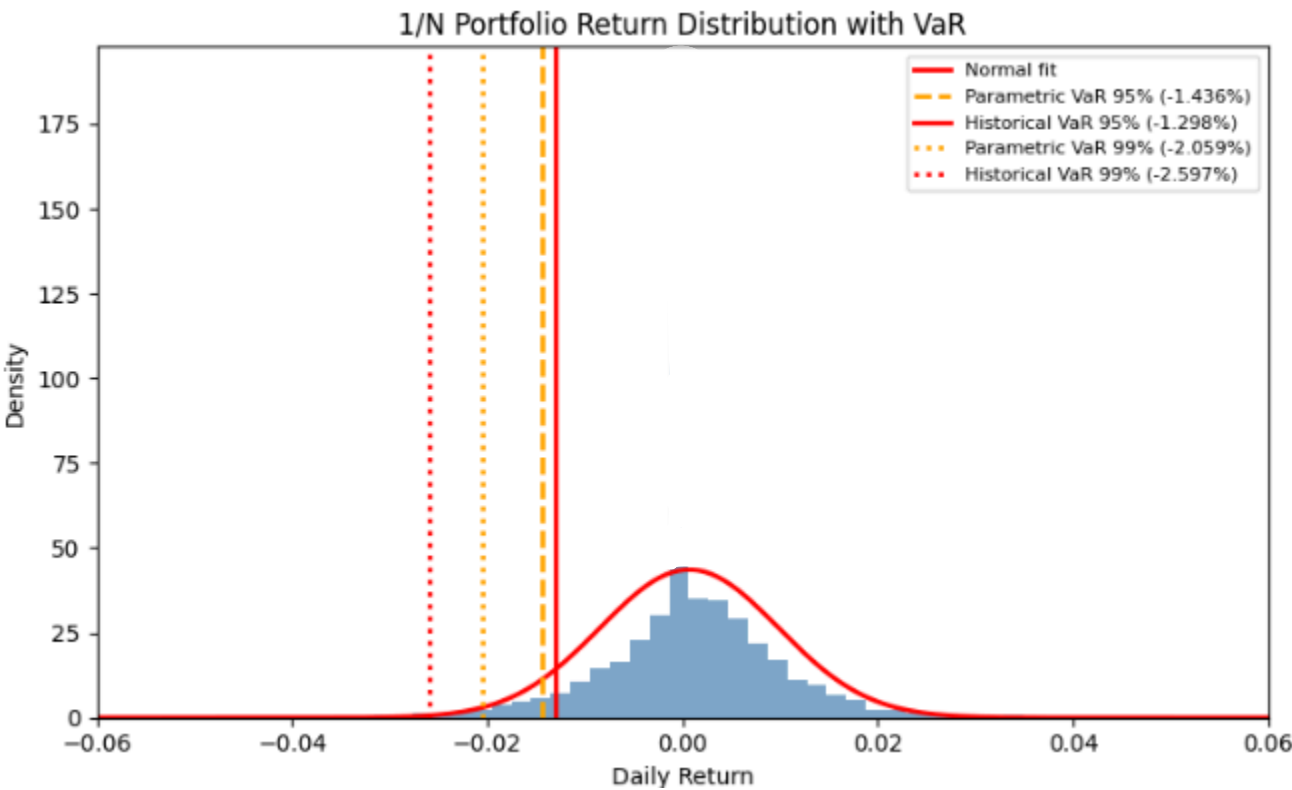
Healthcare

LOWEST VOL



Utilities

FIG 04 • 1/N PORTFOLIO VAR



Parametric VaR understates risk. Historical VaR & Expected Shortfall come in materially more negative.

PARAM VAR 95%

-1.44%

HIST VAR 99%

-2.60%

SECTION

03



Portfolio *optimising*

*Before portfolio construction, optimal strategies
must be identified*

Complexity vs *Simplicity*

■ Naive

- **1/N Equal Weight**
- Random Search

■ Classical

- **Markowitz Max Sharpe**
- Min CVAR
- **Min Variance**
- Moving Average
- Momentum

■ Modern risk Structure

- **Risk Parity**
- **Hierarchal Risk Parity**
- Max Omega
- Max Rachev
- Max Sortino
- Min CDAR
- Max Diversification

■ ML/Advanced

- **LSTM**
- **Dynamic XGBoost**
- Min EVAR
- Min EDAR
- ERO (Novel)



Comparing Complexity *with* Simplicity

Three weight conventions *we tested*.



A

STATIC ABSOLUTE

Fix quantities. Don't touch.

- Quantity fixed at start & does not rebalance
- As prices drift, exposure drifts with them.
- Lower turnover, lower transaction cost.



B

STATIC RELATIVE

Fix proportions. Rebalance.

- Fixed proportions of portfolio & **rebalance over time**
- Continuous rebalancing → systematic "**buy low, sell high.**"
- Slight Sharpe edge: **1.46 vs 1.36.**



C

DYNAMIC

Recompute weights monthly.

- Weights = function of recent stock performance.
- LSTM & XGBoost predict next-month returns.
- **Highest sophistication; highest transaction cost.**



Trained from 1993 - 2016 & Unseen data 2017 - 2019

Static *weights*.

Optimise once on training data, then **hold the weights fixed** through the test period.



Markowitz Max-Sharpe
Best risk-adjusted return



Markowitz Minimum Variance
Lowest risk portfolio

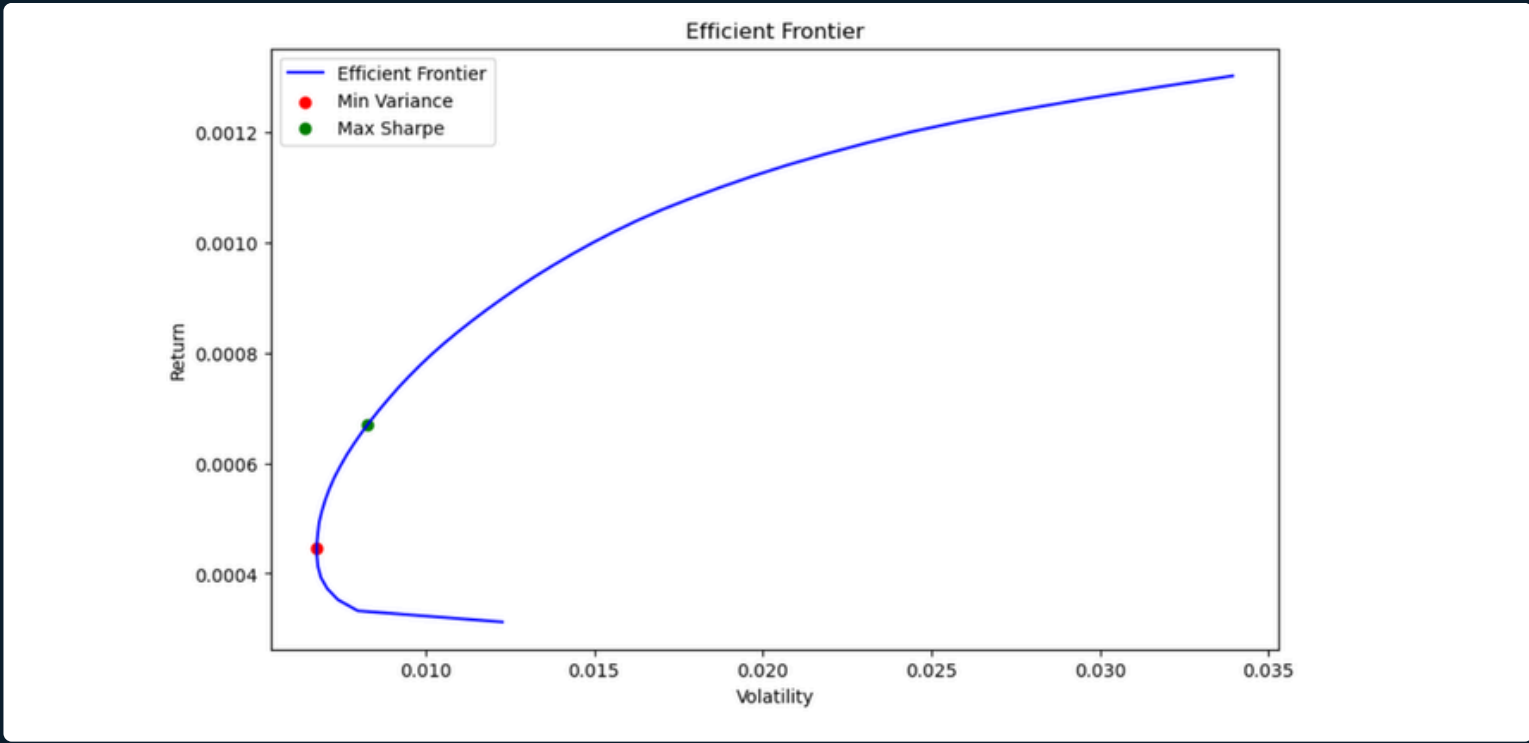


HRP/ Risk Parity
Risk allocation strategies

EFFICIENT FRONTIER

Optimal trade-off between risk and return.

FIG 05 · EFFICIENT FRONTIER



● Max Sharpe = best risk-adjusted

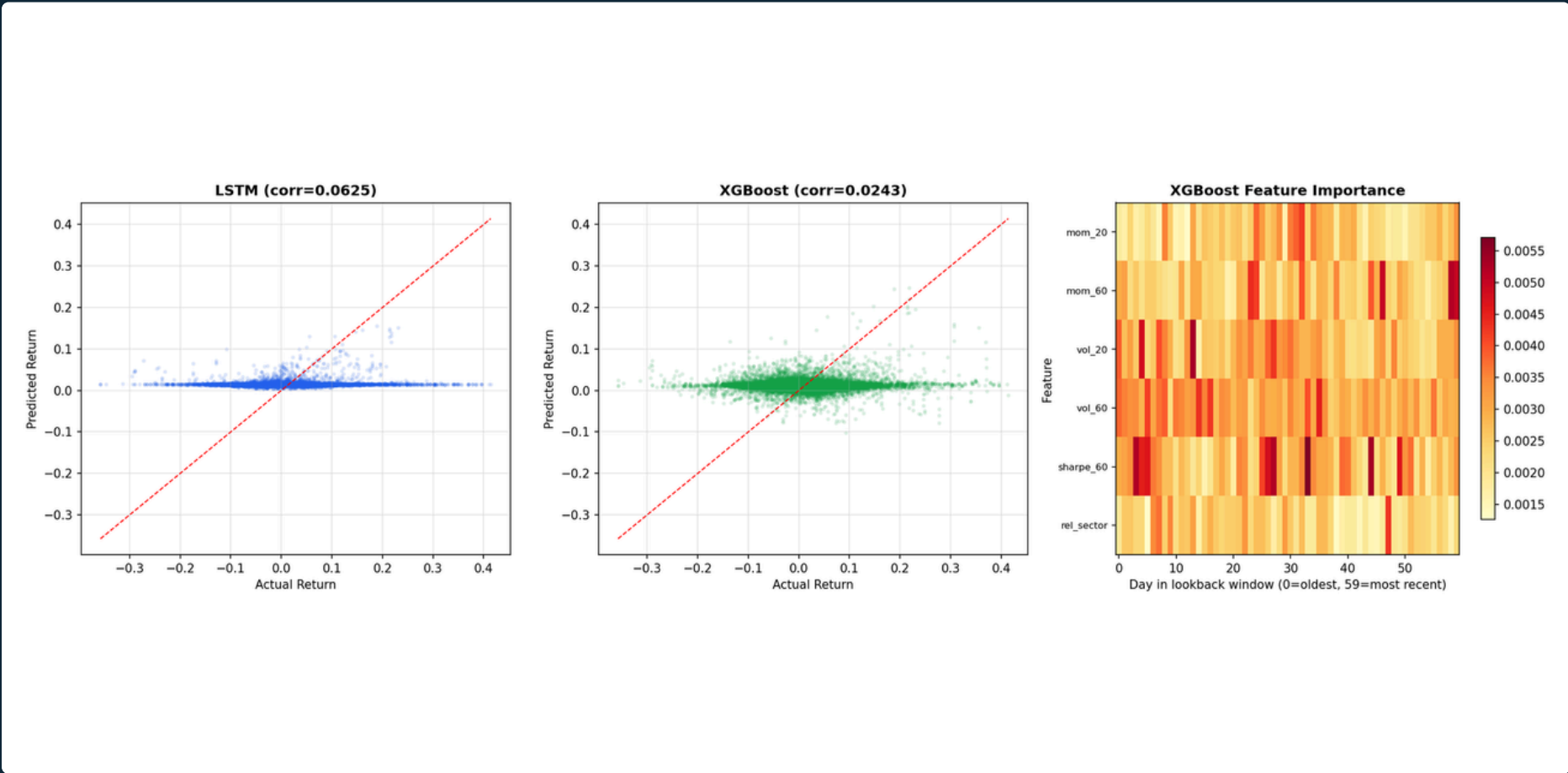
● Min Variance = lowest risk

Dynamic *weights*.

Predict next-month returns with LSTM or XGBoost. Rebalance monthly.



FIG 06 · PREDICTED VS ACTUAL RETURNS & FEATURE IMPORTANCE



LSTM corr
0.0625

XGBoost corr
0.0243

MODEL A



LSTM

Recurrent neural net **sequence model** on engineered features.

MODEL B



XGBoost

Gradient-boosted **decision trees** on the same features.

DIEBOLD-MARIANO TEST

LSTM's MSE is **statistically lower** than XGBoost's but it didn't translate into portfolio wins.



Hypothesis: if anything beats naive diversification, this should.

ERO – an attempt at our own *objective function*.

What if one objective could capture tail loss, upside asymmetry, and distribution shift simultaneously?

$$ERO(w) = \text{Weight} \cdot EVaR(-r^T w) - \text{Weight} \cdot \Omega_L(r^T w) + \text{Weight} \cdot W\text{-penalty}(w)$$

 λ_1

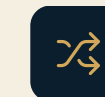
EVaR

Entropic tail loss (Ahmadi-Javid 2012) - strictly tighter than CVaR.

 λ_2

Omega

Upside-to-downside ratio (Keating & Shadwick 2002).

 λ_3

Wasserstein DRO

Distributional robustness - penalty against distribution shift.



RESULT

ERO did **not** beat its components OOS. ERO Sharpe **1.37** vs Min EVaR **1.41**. Composite averaged out individual strengths.



WHY IT STILL MATTERS

Composing risk measures is a valid research direction. The failure points to the weight **tuning**, not a different formula - next step is to optimise the weights.

04

SECTION



Results & *recommendations.*

What we found, whether the wins are real, and what we'd advise the client.

Final in-sample *cumulative results.*

In-sample winners often **don't survive** the out-of-sample test.



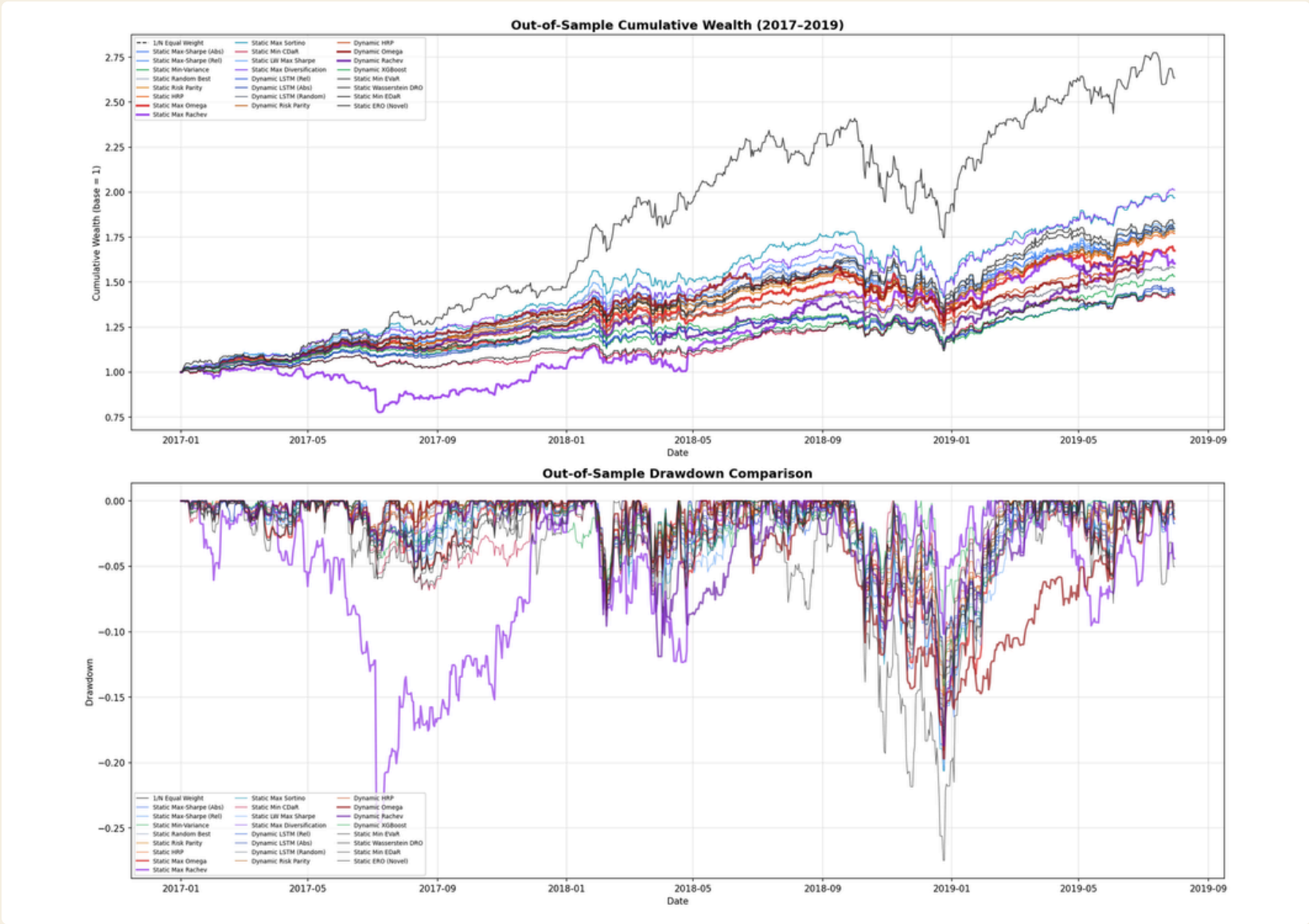
STRATEGY	TYPE OF WEIGHT	FINAL CUMULATIVE VALUE	SHARPE RATIO
Max Sortino	RELATIVE	\$111,687,145	1.47
Max Sharpe	RELATIVE	\$40,713,826	1.28
Momentum	RELATIVE	\$14,383,403	1.18
Min Variance	RELATIVE	\$5,465,255	1.04
Moving Average	RELATIVE	\$3,338,683	1.16

Out-of-sample *results*.

Risk-structure methods dominate. 1/N beat every ML strategy.



FIG 07 · OOS CUMULATIVE WEALTH & DRAWDOWN · 2017-2019



All 25 strategies overlaid. Wasserstein DRO wins on terminal value but with high drawdown; Max Diversification wins on Sharpe.

TOP 5 BY SHARPE

STRATEGY	TYPE	FINAL VALUE	SHARPE
✂ Max Diversification	RELATIVE	\$201,221	1.73
📊 Static HRP	RELATIVE	\$177,009	1.71
🔄 Dynamic Risk Parity	DYNAMIC	\$174,370	1.69
⚖ Static Risk Parity	RELATIVE	\$178,309	1.64
÷ 1/N	STATIC	\$179,424	1.62



The naive 1/N baseline beat every single machine-learning strategy out-of-sample.

Limitations & three *takeaways*.

LIMITATIONS

-  2.5-year OOS window: too short for significance.
-  No transaction costs
-  Some risk measures still assume Gaussian tails.
-  Single backtest path - no cross-validation across windows.
-  US equities only - no international, fixed-income, or factors.

THREE TAKEAWAYS

1 Structure beat prediction.
Diversification structure won across 25 strategies.

2 Naive 1/N beat every ML strategy.
Out-of-sample, the dumbest baseline outperformed.

3 Rankings real, significance not.
Needs 10–15 years of OOS data to confirm.

Recommendations to *future clients*.



DECISION-MAKING

Default to 1/N. Deviate only with reason.

- ✓ Default to **1/N**. Only deviate when justified.
- ✓ Prefer **risk-structure** over return-prediction.
- ✓ Quarterly rebalancing **minimum** (transaction costs).



FUTURE ANALYSIS

Where the research goes next.

- **Walk-forward validation** across market regimes.
- **Ensemble ML** models, not single estimators.
- Explicit **transaction-cost** modelling.

Thank you.

ALPAY • ROHAN • BYRON • TYMON • YALINDEE